

ОТЧЁТ О ВЫПОЛНЕНИИ ЛАБОРАТОРНОЙ РАБОТЫ №2

НАСТРОЙКА ПРОТОКОЛА STP (IEEE 802.1D)

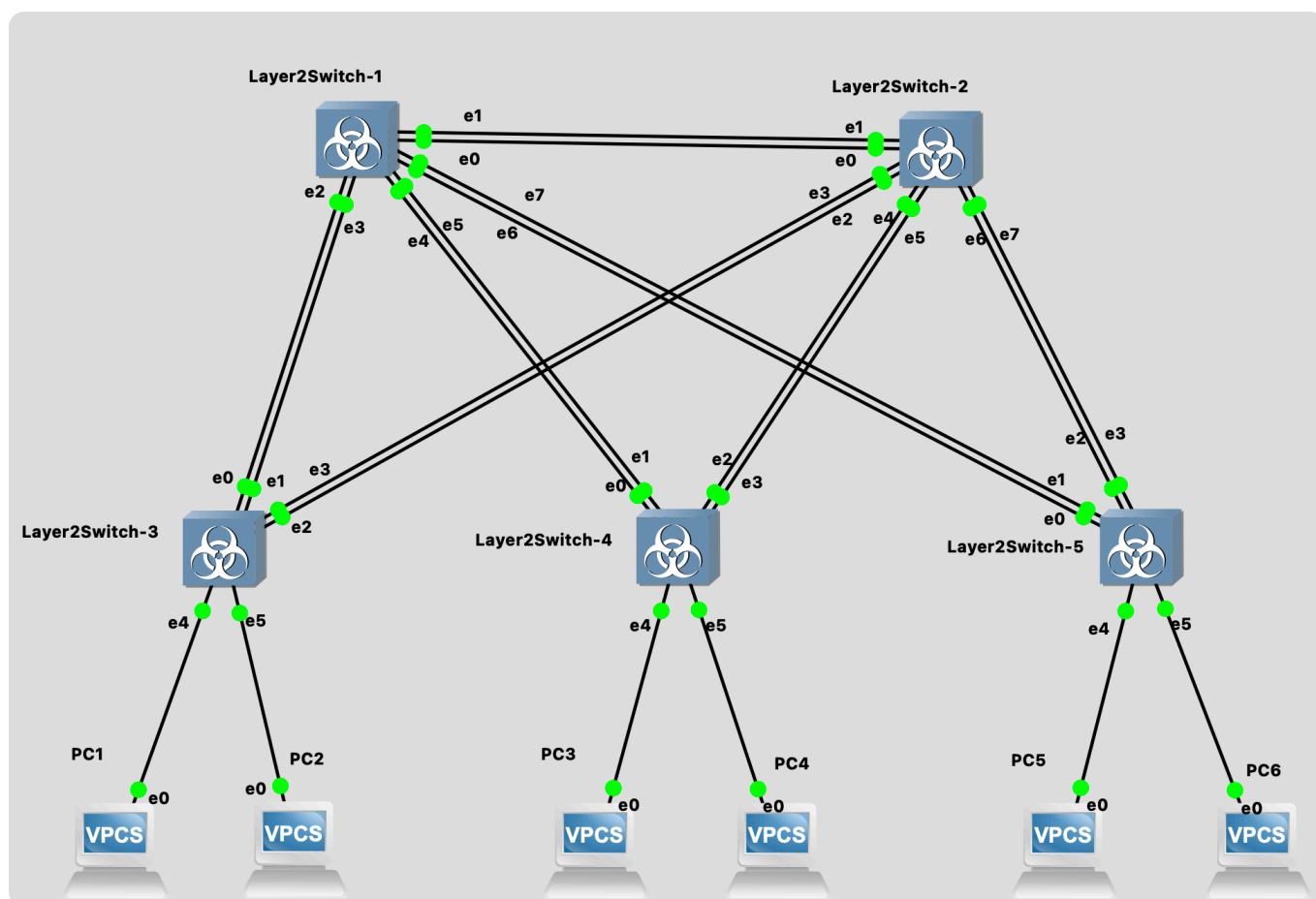
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Цель работы

Настройка протокола Spanning Tree Protocol (STP, IEEE 802.1D) на коммутаторах Cisco IOSv-L2, выбор корневого коммутатора путем изменения приоритета, проверка связности сети между всеми узлами VPCS, фиксация топологии и состояний портов, анализ BPDU пакетов в Wireshark и исследование перестроения топологии при изменении стоимости маршрута.

1. Топология сети и назначение корневого коммутатора (Root Bridge)

Исходная топология сети состоит из 5 коммутаторов Cisco (S1–S5) и 6 персональных компьютеров (PC1–PC6), находящихся в общей подсети 192.168.1.0/24 (VLAN 1).



Шаг 1. Назначение S1 корневым коммутатором

По умолчанию на всех коммутаторах установлен приоритет 32768 (с учетом расширенного системного номера VLAN 1 — 32769). Для гарантированного назначения коммутатора S1 (Layer2Switch-1) в качестве корневого моста (Root Bridge) для VLAN 1 устанавливается пониженный приоритет 4096:

```
S1> enable
S1# configure terminal
S1(config)# hostname S1
S1(config)# spanning-tree vlan 1 priority 4096
S1(config)# exit
S1# write
```

Шаг 2. Проверка статуса STP на корневом коммутаторе S1

```
S1# show spanning-tree vlan 1
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    4097
```

```
Address    0cab.64a6.0000
```

```
This bridge is the root
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority    4097 (priority 4096 sys-id-ext 1)
```

```
Address    0cab.64a6.0000
```

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr
Gi1/2	Desg	FWD	4	128.7	Shr
Gi1/3	Desg	FWD	4	128.8	Shr
Gi2/0	Desg	FWD	4	128.9	Shr

Вывод: Коммутатор S1 стал корневым (This bridge is the root), все его порты перешли в состояние **Designated Forwarding (Desg FWD)** со стоимостью 4 (GigabitEthernet).

2. Проверка связности всех хостов

Адресация хостов:

- **PC1:** 192.168.1.10/24
- **PC2:** 192.168.1.20/24
- **PC3:** 192.168.1.30/24
- **PC4:** 192.168.1.40/24
- **PC5:** 192.168.1.50/24
- **PC6:** 192.168.1.60/24

Шаг 1. Проверка с PC1 (192.168.1.10)

```
PC1> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=2.005 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=64 time=1.251 ms

PC1> ping 192.168.1.30
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=16.422 ms
84 bytes from 192.168.1.30 icmp_seq=2 ttl=64 time=7.116 ms

PC1> ping 192.168.1.40
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=14.360 ms

PC1> ping 192.168.1.50
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=13.412 ms

PC1> ping 192.168.1.60
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=18.527 ms
```

Шаг 2. Проверка с PC2 (192.168.1.20)

```
PC2> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=6.458 ms

PC2> ping 192.168.1.30
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=15.452 ms

PC2> ping 192.168.1.40
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=17.209 ms

PC2> ping 192.168.1.50
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=8.879 ms

PC2> ping 192.168.1.60
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=15.042 ms
```

Шаг 3. Проверка с PC3 (192.168.1.30)

```
PC3> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=13.901 ms

PC3> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=17.163 ms

PC3> ping 192.168.1.40
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=11.251 ms

PC3> ping 192.168.1.50
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=9.035 ms

PC3> ping 192.168.1.60
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=18.699 ms
```

Шаг 4. Проверка с PC4 (192.168.1.40)

```
PC4> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=13.019 ms

PC4> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=14.397 ms

PC4> ping 192.168.1.30
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=6.871 ms

PC4> ping 192.168.1.50
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=11.180 ms

PC4> ping 192.168.1.60
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=10.915 ms
```

Шаг 5. Проверка с PC5 (192.168.1.50)

```
PC5> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=11.443 ms

PC5> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=12.768 ms

PC5> ping 192.168.1.30
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=20.640 ms

PC5> ping 192.168.1.40
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=27.711 ms

PC5> ping 192.168.1.60
84 bytes from 192.168.1.60 icmp_seq=1 ttl=64 time=4.325 ms
```

Шаг 6. Проверка с PC6 (192.168.1.60)

```
PC6> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=17.679 ms

PC6> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=14.759 ms

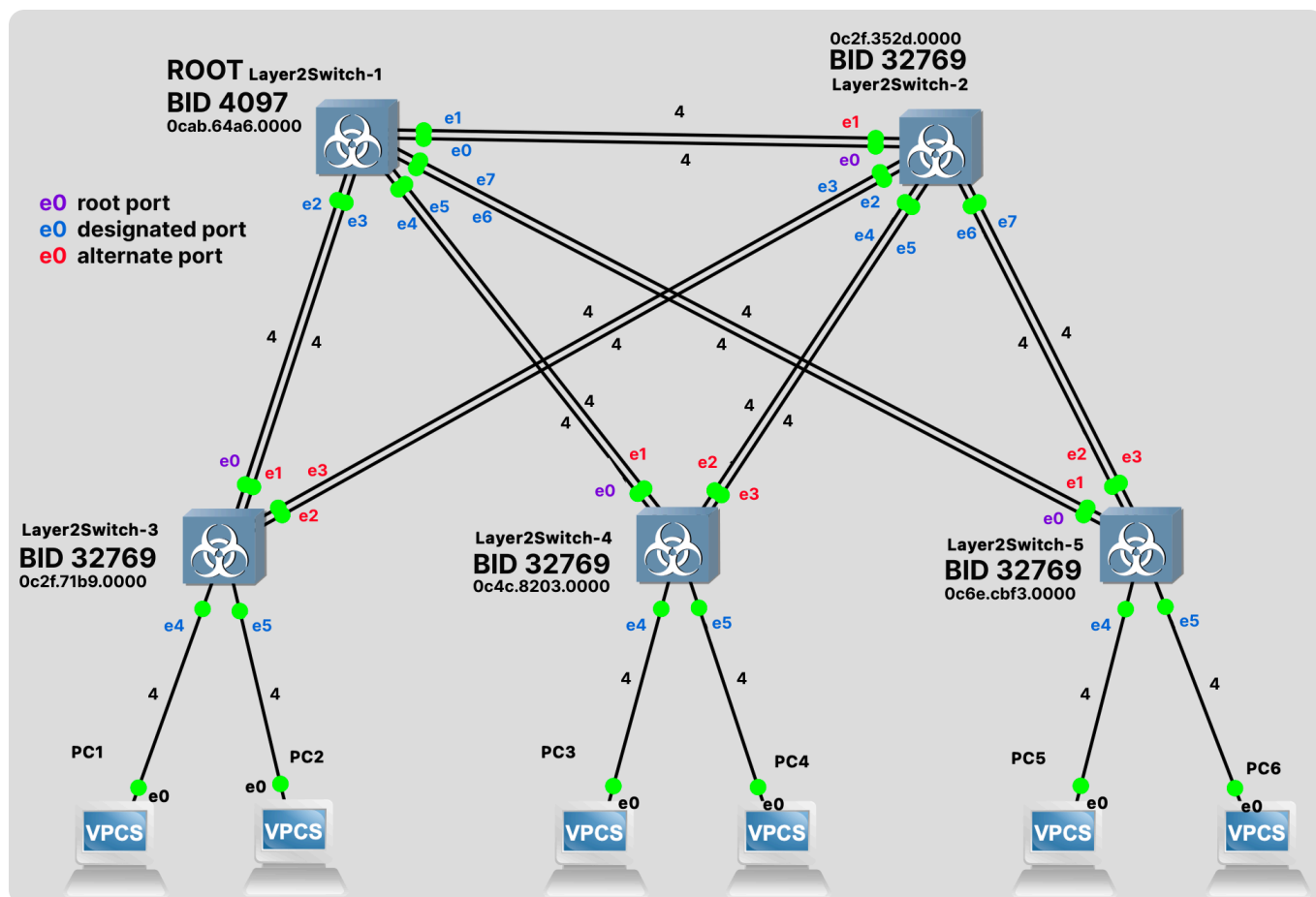
PC6> ping 192.168.1.30
84 bytes from 192.168.1.30 icmp_seq=1 ttl=64 time=7.081 ms

PC6> ping 192.168.1.40
84 bytes from 192.168.1.40 icmp_seq=1 ttl=64 time=6.296 ms

PC6> ping 192.168.1.50
84 bytes from 192.168.1.50 icmp_seq=1 ttl=64 time=6.743 ms
```

Результат: Все узлы имеют 100% связность, петли в сети заблокированы STP.

3. Таблица BID и состояний портов коммутаторов



4. Захват Hello BPDU через Wireshark со всех линков Root Bridge

Layer2Switch-1_Ethernet0_to_Layer2Switch-2_Ethernet0

Spanning Tree Protocol

```
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
> BPDU flags: 0x00
√ Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Root Bridge Priority: 4096
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
    Root Path Cost: 0
√ Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Bridge Priority: 4096
    Bridge System ID Extension: 1
    Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
    Port identifier: 0x8001
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
√ Originating VLAN (PVID): 1
    Type: Originating VLAN (0x0000)
    Length: 2
    Originating VLAN: 1
```

Spanning Tree Protocol

```
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00) <- Cam Hello BPDU
BPDU flags: 0x00
Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Root Bridge Priority: 4096
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0 <- Стоимость 0 потому, что это и так корневой свитч
Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Bridge Priority: 4096
    Bridge System ID Extension: 1
    Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8001 <- Порт, с которого S1 отправил пакет
Message Age: 0
Max Age: 20
Hello Time: 2 <- Интервал отправки Hello BPDU
Forward Delay: 15
Originating VLAN (PVID): 1
    Type: Originating VLAN (0x0000)
    Length: 2
    Originating VLAN: 1
```

Layer2Switch-1_Ethernet1_to_Layer2Switch-2_Ethernet1

Spanning Tree Protocol

```
Protocol Identifier: Spanning Tree Protocol (0x0000)
```

```
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Root Bridge Priority: 4096
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0
Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Bridge Priority: 4096
    Bridge System ID Extension: 1
    Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8002
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
Originating VLAN (PVID): 1
```

Layer2Switch-1_Ethernet2_to_Layer2Switch-3_Ethernet0

Spanning Tree Protocol

```
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Root Bridge Priority: 4096
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0
Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Bridge Priority: 4096
    Bridge System ID Extension: 1
    Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8003
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
Originating VLAN (PVID): 1
```

Layer2Switch-1_Ethernet3_to_Layer2Switch-3_Ethernet1

Spanning Tree Protocol

```
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
    Root Bridge Priority: 4096
    Root Bridge System ID Extension: 1
    Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0
```

```
Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
  Bridge Priority: 4096
  Bridge System ID Extension: 1
  Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8004
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
Originating VLAN (PVID): 1
```

Layer2Switch-1_Ethernet4_to_Layer2Switch-4_Ethernet0

```
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
  Root Bridge Priority: 4096
  Root Bridge System ID Extension: 1
  Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0
Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
  Bridge Priority: 4096
  Bridge System ID Extension: 1
  Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8005
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
Originating VLAN (PVID): 1
```

Layer2Switch-1_Ethernet5_to_Layer2Switch-4_Ethernet1

```
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
  Root Bridge Priority: 4096
  Root Bridge System ID Extension: 1
  Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0
Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
  Bridge Priority: 4096
  Bridge System ID Extension: 1
  Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8006
Message Age: 0
Max Age: 20
Hello Time: 2
```

Forward Delay: 15
Originating VLAN (PVID): 1

Layer2Switch-1_Ethernet6_to_Layer2Switch-5_Ethernet0

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
 Root Bridge Priority: 4096
 Root Bridge System ID Extension: 1
 Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0
Bridge Identifier: 4096 / 1 / 0c:ab:64:a6:00:00
 Bridge Priority: 4096
 Bridge System ID Extension: 1
 Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8007
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
Originating VLAN (PVID): 1

Layer2Switch-1_Ethernet7_to_Layer2Switch-5_Ethernet1

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 24576 / 100 / 0c:ab:64:a6:00:00
 Root Bridge Priority: 24576
 Root Bridge System ID Extension: 100
 Root Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Root Path Cost: 0
Bridge Identifier: 24576 / 100 / 0c:ab:64:a6:00:00
 Bridge Priority: 24576
 Bridge System ID Extension: 100
 Bridge System ID: 0c:ab:64:a6:00:00 (0c:ab:64:a6:00:00)
Port identifier: 0x8008
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
Originating VLAN (PVID): 100
 Type: Originating VLAN (0x0000)
 Length: 2
 Originating VLAN: 100

5. Изменение стоимости Root Port у некорневого коммутатора

Сделаю стоимость пути через корневой порт у S3 равной 100:

```
S3#configure terminal
S3(config)#interface gigabitEthernet 0/0
S3(config-if)#spanning-tree vlan 1 cost 100
S3(config-if)#end
S3#show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

Root ID	Priority	4097
	Address	0cab.64a6.0000
	Cost	4
	Port	2 (GigabitEthernet0/1)
	Hello Time	2 sec
	Max Age	20 sec
	Forward Delay	15 sec

Bridge ID	Priority	32769 (priority 32768 sys-id-ext 1)
	Address	0c2f.71b9.0000
	Hello Time	2 sec
	Max Age	20 sec
	Forward Delay	15 sec
	Aging Time	15 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Altn	BLK	100	128.1	Shr
Gi0/1	Root	LIS	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

После смены Root Port состояние порта становится LIS (listening), через 15 секунд оно станет LRN (learning), через ещё 15 секунд будет FWD (forwarding).

Gi0/1	Root	FWD	4	128.2	Shr
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В этом примере связь между S3 и S1 всё ещё прямая, поэтому Root Path Cost равен 4.

Заблокирую второй порт до S1, чтобы трафик шёл с S3 до S1 через S2:

```
S3#configure terminal
S3(config)#interface gigabitEthernet 0/1
S3(config-if)#spanning-tree vlan 1 cost 100
S3(config-if)#end
S3#show spanning-tree vlan 1
```

VLAN0001

Spanning tree enabled protocol ieee

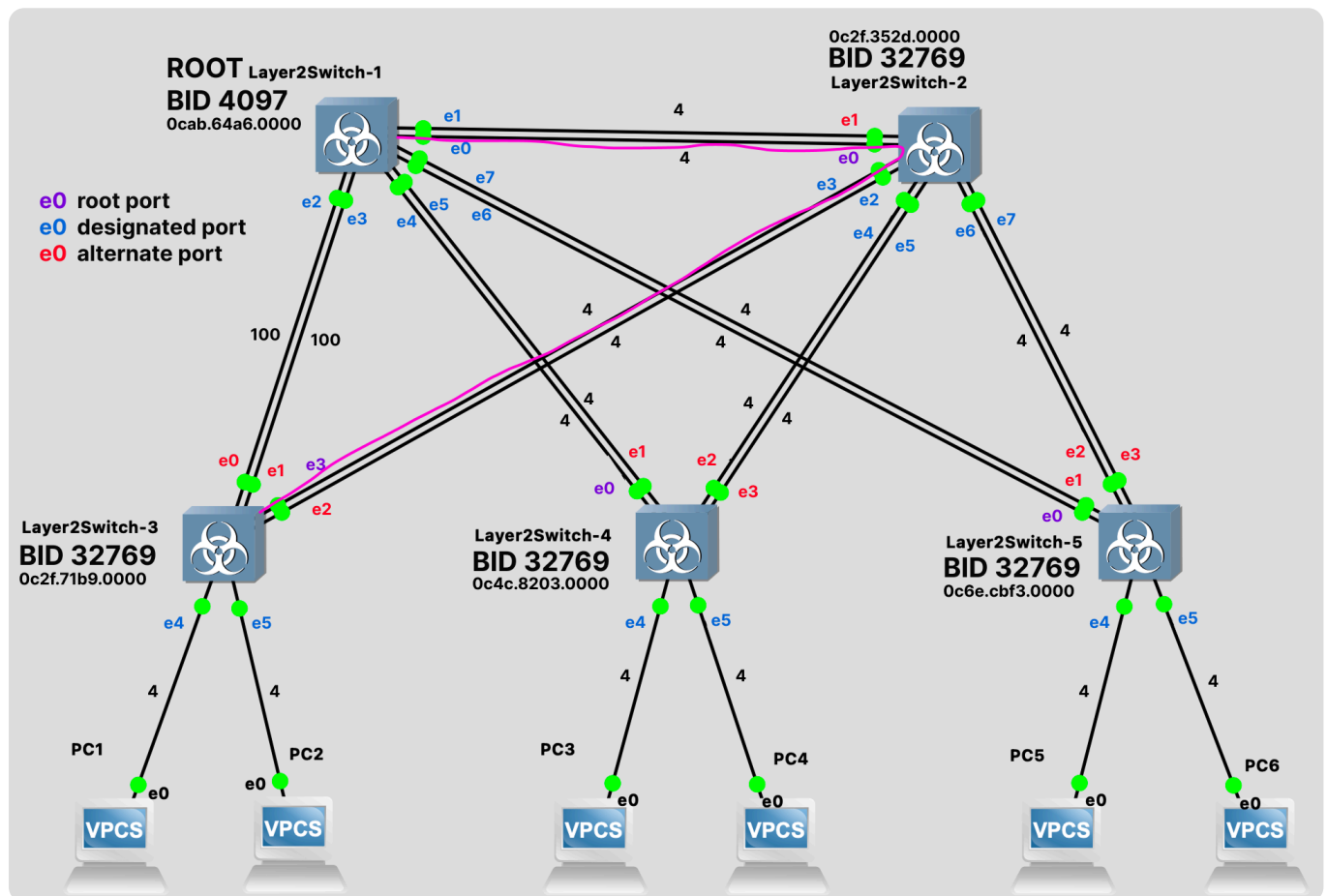
Root ID	Priority	4097
	Address	0cab.64a6.0000
	Cost	8
	Port	3 (GigabitEthernet0/2)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0c2f.71b9.0000
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 15 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Altn	BLK	100	128.1	Shr
Gi0/1	Altn	BLK	100	128.2	Shr
Gi0/2	Root	FWD	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gi1/0	Desg	FWD	4	128.5	Shr
Gi1/1	Desg	FWD	4	128.6	Shr

Теперь Cost равен 8, оба порта в S1 заблокированы, путь трафика изменился и проходит через S2.



6. Конфигурации

Сохранены по пути `./configs`

7. Разница STP и RSTP

RSTP сохраняет ту же логику выбора Root Bridge и ролей портов, что и STP, но устраняет фиксированные таймеры ожидания, заменяя их активным согласованием между соседними коммутаторами, что сокращает время восстановления сети с 30–50 секунд до 1–2 секунд.